SEBASTIÁN FERNÁNDEZ RIVERA · APRIL 27, 2026

Claude Code for Economic Research

Introduction and Setup

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Introduction to Claude Code

What is Claude Code?

DEFINITION

Claude Code is an agentic coding tool developed by Anthropic that operates directly in your terminal, IDE, desktop app, and browser to help turn ideas into code faster than ever before.

Key Capabilities:

- Builds features from plain English descriptions
- Creates plans, writes code, and ensures functionality
- Automates tedious tasks.

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Applications in Economic Research

Research Workflow Support

COMPLETE DATA ANALYSIS PIPELINE

Claude Code can handle the entire research workflow from data cleaning to publication-ready outputs.

Research Capabilities:

1. **Data Management:** Navigate codebases and datasets
2. **Statistical Analysis:** Run econometric models
3. **Visualization:** Generate publication-quality figures and tables
4. **Documentation:** Read and document code for replicability

Econometric Modeling

ADVANCED STATISTICAL METHODS

As a coding tool, Claude Code can assist in the implementation of econometric techniques essential for modern economic research

Claude can be integrated with widely used software:

- Native support for R and Python
- Support for Stata and Matlab via specialized MCP servers

Example: Running a DiD Analysis

NATURAL LANGUAGE TO CODE

You can describe your research design in plain English:

```
$ claude "Run a difference-in-differences regression
on the treatment effect of minimum wage policy using
panel data from 2010-2020, controlling for state and
year fixed effects"
```

Claude Code will:

- Load and clean the dataset
- Construct appropriate variables
- Run the regression with robust standard errors
- Generate regression tables and visualizations

Data Science Integration

SEAMLESS WORKFLOW INTEGRATION

Data scientists and economists use Claude Code to streamline their entire analytical pipeline.

Data Operations

- Write SQL queries
- Clean messy datasets
- Transform variables
- Merge data sources

Analysis & Output

- Create visualizations
- Generate summary statistics
- Export results
- Document methods

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Advanced Features for Research

Model Context Protocol (MCP)

EXTENSIBILITY THROUGH MCP SERVERS

Claude Code can connect to external data sources and tools through MCP servers, enabling access to proprietary datasets and specialized tools.

Integration Examples:

- Google Drive for institutional data repositories
- GitHub for version control and collaboration
- Specialized databases (WRDS, Bloomberg, FRED)
- Cloud storage services (AWS S3, Azure)
- Custom APIs for proprietary data feeds

Project-Wide Awareness

INTELLIGENT CONTEXT MANAGEMENT

Claude Code maintains awareness of your entire research project structure and dependencies.

Benefits:

1. Understands relationships between scripts and datasets
2. Identifies relevant files for specific analytical tasks
3. Tracks variable definitions across multiple files
4. Maintains consistency in naming conventions
5. Documents code dependencies automatically

Reproducibility & Transparency

BEST PRACTICES BUILT-IN

Claude Code encourages reproducible research through transparent code generation and comprehensive documentation.

Reproducibility Features:

- Version control: Automatic Git integration
- Documentation: Self-documenting code with clear comments
- Environment management: Consistent dependency tracking
- Audit trails: Complete record of analytical decisions
- Replication packages: Easy creation of replication archives

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Installation Guide

Installation Overview

MULTI-PLATFORM SUPPORT

Claude Code supports macOS, Windows, and Linux with multiple installation methods for each platform.

Key Points:

- Native installers require no Node.js dependency
- Auto-updates keep you on the latest version (native installs)
- Package managers available for all major platforms
- One-time authentication via OAuth

macOS Installation

NATIVE INSTALLER FOR MACOS

The simplest and most reliable installation method.

```
# Open Terminal (Cmd + Space, type "Terminal")

# Run installation script
curl -fsSL https://claude.ai/install.sh | bash

# Verify installation
claude --version
```

Windows Installation

POWERSHELL INSTALLER FOR WINDOWS

Native Windows installation.

```
# Open PowerShell

# Run installation
irm https://claude.ai/install.ps1 | iex

# Install Git (required)
# Download from gitforwindows.org

# Verify installation
claude --version
```

Linux Installation

UNIVERSAL INSTALLATION

Single command installation for all major distributions.

```
# Open terminal (usually Ctrl+Alt+T)

# Run installation script
curl -fsSL https://claude.ai/install.sh | bash

# Verify installation
claude --version
```

Authentication Setup

ONE-TIME OAUTH PROCESS

After installation, authenticate with your Claude account.

```
# Start Claude Code
claude

# When prompted:
# - Choose "Claude account with subscription"
# - NOT "Anthropic Console"
# - Complete browser authentication
```

Requirements:

- Claude Pro, Max or Anthropic Console account
- Active subscription
- Web browser for OAuth completion

Verifying Installation

HEALTH CHECK

Use the built-in diagnostic tool to verify your setup.

```
# Inside Claude Code, run:
/doctor
```

What `/doctor` Checks:

- Installation integrity
- Active subscription status
- System configuration
- Common setup issues

Healthy output shows green checkmarks for all items.

Installation Best Practices

IMPORTANT REMINDERS

Follow these guidelines for a smooth installation experience.

- **Close terminal after Node.js install:** Refresh system PATH completely
- **Prefer native installers:** Avoid deprecated methods
- **Navigate to project directory:** Start Claude Code from your research folder
- **Keep updated:** Native installs auto-update; package managers require manual updates
- **Check documentation:** Visit `docs.claude.com` for troubleshooting

Conclusion: Why Claude Code for Economics?

KEY ADVANTAGES

Claude Code transforms the economic research workflow by combining speed, accuracy, and reproducibility.

1. **Speed:** Accelerate time-consuming coding tasks
2. **Transparency:** Every step is documented and auditable
3. **Flexibility:** Handles diverse data cleaning and econometric methods
4. **Integration:** Connects to institutional data sources
5. **Reproducibility:** Built-in version control and documentation

IMPORTANT

Claude Code is an assistant, not a replacement for your expertise. Always review all code and results it generates. Final responsibility for accuracy rests with you.

Thank you.

sebasfr.github.io